

Zoe Tian

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Backend and infrastructure engineer with 6+ years of experience building distributed systems, search and retrieval infrastructure, and ML-powered ranking systems. Experienced in designing high-performance serving and data pipelines in Go and Python, with recent work spanning recommendation systems, ML infrastructure, and agent-driven ranking optimization at Meta.

WORK EXPERIENCE

Meta Bellevue, WA
Senior Software Engineer (MRS - Modern Recommendation Systems) April 2025 - Present

- Led cross-team technical design of an **agent-driven GNN ranking policy optimization framework** for Facebook Reels and Deep Dive surface, enabling autonomous policy search across user cohorts and ranking configurations, increasing **video view seconds (VVS)** by **6%** and **like-sum engagement** by **6.37%**
- Owned the end-to-end policy evaluation workflow across 10+ candidate ranking strategies, including seed experiment selection, policy generation, offline metric analysis, and cross-functional launch coordination with ranking, GNN, and video recommendation teams
- Built a full-stack prototype for causal video recommendation, designing scalable candidate retrieval pipelines and contextual recommendation interfaces using **React, TypeScript, Python, and Hack**
- Optimized distributed data preprocessing pipeline for the ranking system, reducing CPU label parsing time by **86% (550ms to 75ms)** and machine resource usage by **25%** through UDF optimization and elimination of redundant computation
- Mentored junior engineers through knowledge sharing, and guidance on team processes and best practice

Dropbox Vancouver, BC
Infrastructure Software Engineer (Search Retrieval & ML team) March 2022 - March 2024

- Designed and implemented Daggo: a 1.5-year project that transform query serving paths into a DAG structure in **Golang**
- Reduced **P95 Search latency** by **200ms** through the application and deployment of Daggo
- Set up **experiments** to safely verify the latency gain in each DAG-ified components with live shadow traffic
- Created a visualization tool in **D3.js** for debugging and troubleshooting DAG execution latency
- Repeatedly analyzed critical path with **Hive** and **DAG logs**, modified framework and call paths accordingly, until hitting latency goal
- Improved paid enterprise user experience by fixing bugs in search with the eastern locales, and search by tags in Japanese during on-call rotations

VMware Toronto, ON
Member of Technical Staff II (Tanzu Application Platform) April 2020 - March 2022

- Led the integration of **Backstage.io** as the **developer portal foundation**, reducing new service onboarding time by 70%
- Built automated service templates in Backstage.io, reducing microservice creation time by 60%
- Created **Backstage plugins** for VMware tools integration, serving 1000+ services
- Onboarded two new members onto the team, mentored junior engineer, and facilitate internal meetings

Software Engineer (Concourse CI team) June 2019 - April 2020

- Shipped new features for Concourse CI in **Golang and Elm** followed strict TDD and Agile
- Scaled up the previous backend **authentication** flow for super admin users with large pipelines redesigns
- Re-wrote dashboard endpoints in **Elm** and improved the performance by 50% when loading complex responses
- Worked in a multi-cloud environment with **AWS, GCP, VSphere**; developed with **Docker, Kubernetes, Terraform**

EDUCATION

University of Waterloo Waterloo, ON
Bachelor of Computer Science, Minor in Combinatorics & Optimization 2015 - 2019

PROJECTS

Midi Melody github.com/zoetian/midi_melody

YouTube Music Video Converter in Python

- Applied Youtube-dl library to convert random audio files into mp3
- Extract the melody from an audio file and convert it to MIDI notes
- Output piano keyboard canvas animation as learning tutorials with sheet script pdf

Docker Filter github.com/zoetian/docker-filter

Tiny Image Processing Experiment

- Built a Serverless function to do style transfers with Docker, Openfaas, Tensorflow
- Supported 40+ different styles from Magenta's models

Bustub - CMU Database 445/645 github.com/zoetian/bustub

A relational database management system

- Built a disk-oriented database management system in C++ to support basic SQL